

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Fitness Buddy Project

Project Tile – Fitness Buddy

Domain of Project – Healthcare and Fitness

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Samarth Kadam		202401040131@mitaoe.ac.in	8080126820

Problem Statement :

In today's fast-paced world, maintaining a healthy lifestyle is increasingly difficult. Existing fitness solutions are expensive, rigid, and impersonal. There is a critical need for an intelligent, accessible, and personalized virtual fitness assistant available to everyone — anytime, anywhere.

- **High Cost of Fitness Guidance** - Personal trainers, gym memberships, and nutrition consultations are expensive and inaccessible for most people.
- **Time Constraints & Rigid Schedules** - Traditional fitness programs demand fixed schedules that don't fit busy lifestyles, desk jobs, or family responsibilities.
- **Lack of Personalization** - Generic workout and diet plans ignore individual body stats, goals, dietary preferences, and cultural food habits.
- **Inconsistent Motivation** - Without ongoing support and encouragement, most people abandon their fitness goals within weeks of starting.
- **No Culturally Relevant Nutrition Guidance** - Most fitness apps ignore regional dietary habits. Indian users lack guidance tailored to dal, roti, sabzi, and traditional meal patterns.

Proposed Solution

Fitness Buddy AI is an AI-powered web application built using Python Flask and IBM Watsonx.ai (Granite model). It provides personalized fitness guidance, workout plans, meal recommendations, and health tracking through an interactive chatbot.

Key Features

- AI Fitness Chatbot – Answers fitness and nutrition queries in real time.
- Personalized Workout Plans – Beginner, Intermediate, Weight Loss, and Yoga plans.
- Indian Meal Plans – Diet plans for Weight Loss, Muscle Gain, and Balanced Fitness.
- BMI & Calorie Calculator – Calculates BMI, calorie needs, and ideal weight range.
- Habit Tracker – Tracks water intake, sleep, workouts, and daily steps.
- Daily Motivation – Provides fitness tips and motivational quotes to keep users consistent.

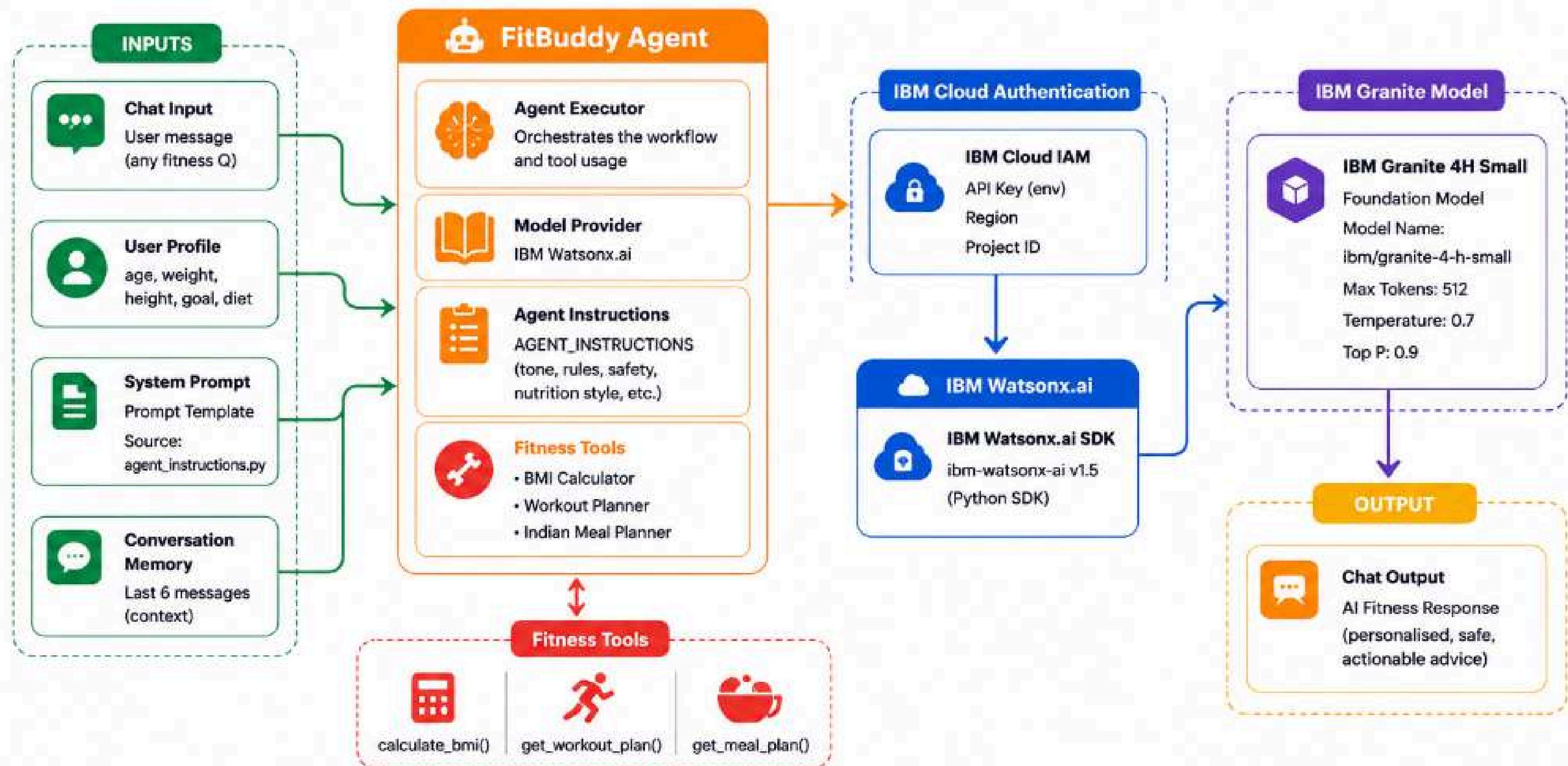
Technology Used

Technology	Purpose
IBM Cloud Lite	API key authentication and cloud services
IBM Watsonx.ai	AI platform for hosting and accessing Granite models
IBM Granite 4H Small	Generates AI-based fitness and nutrition responses
Python Flask	Backend development, REST APIs, and session management
IBM Watsonx.ai SDK	Connects Flask application with Watsonx.ai
Bootstrap 5 & CSS	Responsive, mobile-friendly user interface
JavaScript	Chatbot interaction, BMI calculator, workout timer, and habit tracker

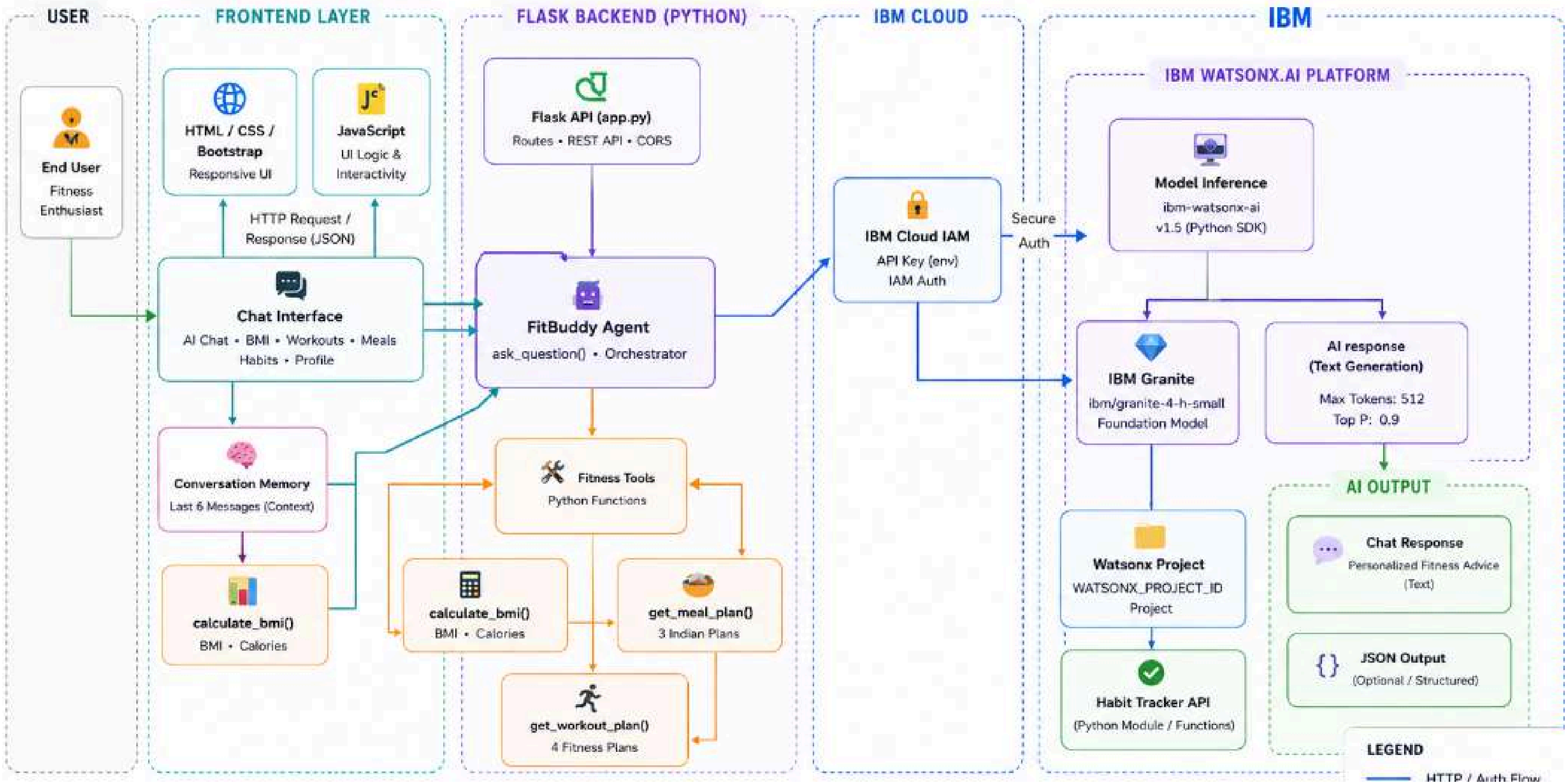
Langflow component Used -

- **Chat Input** – Receives user fitness queries.
- **Data Input (User Profile)** – Collects age, height, weight, fitness goal, and diet preference.
- **Prompt Template** – Loads system instructions and AI behavior.
- **Conversation Memory** – Maintains chat history for context-aware responses.
- **Agent Executor (FitBuddy Agent)** – Processes requests and manages workflow.
- **Tool Node** – Executes BMI calculation, workout planning, and meal planning.
- **IBM Cloud IAM** – Authenticates using IBM Cloud API Key.
- **IBM Watsonx.ai** – Connects the application to IBM AI services.
- **IBM Granite 4H Small** – Generates personalized AI fitness responses.
- **Chat Output** – Displays the final response to the user.

Langflow workflow -



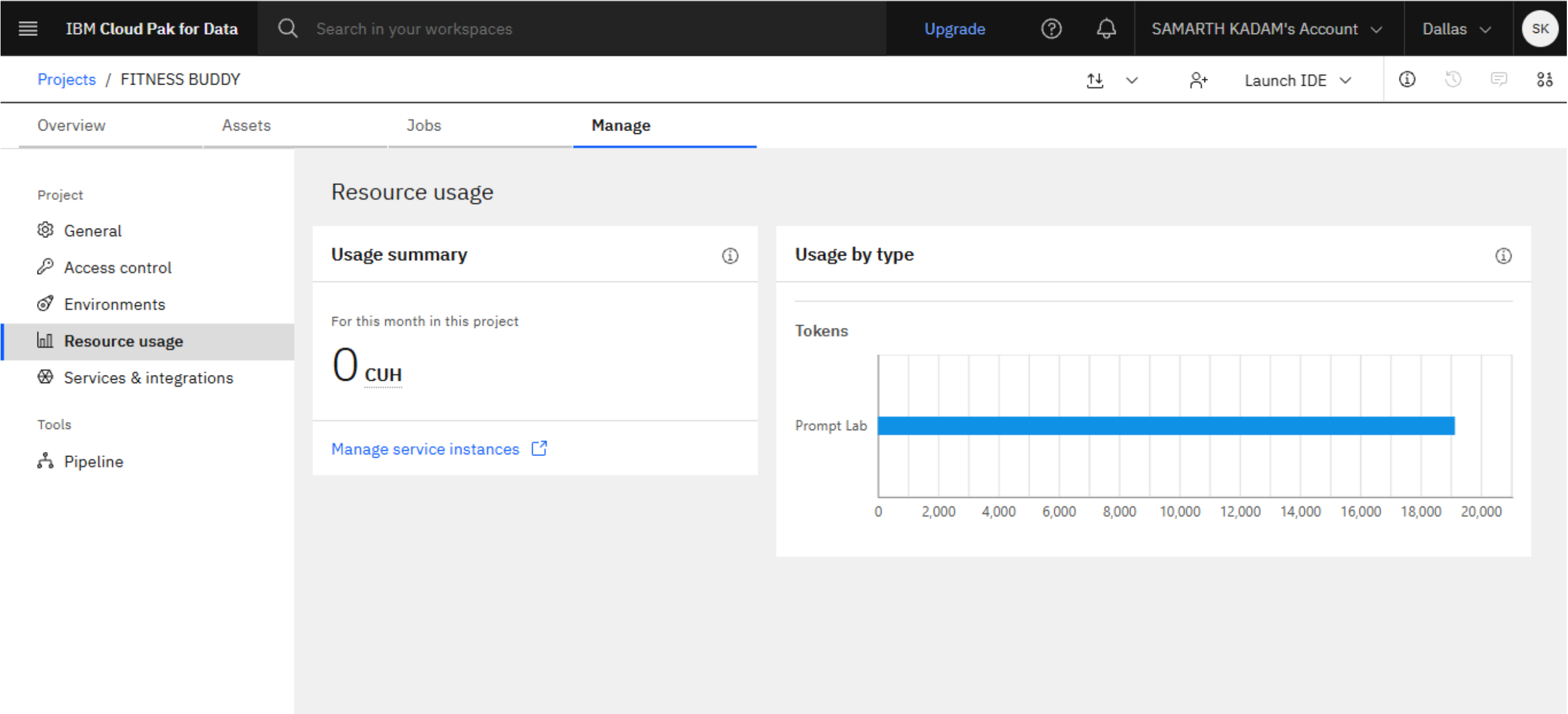
Architecture blueprint



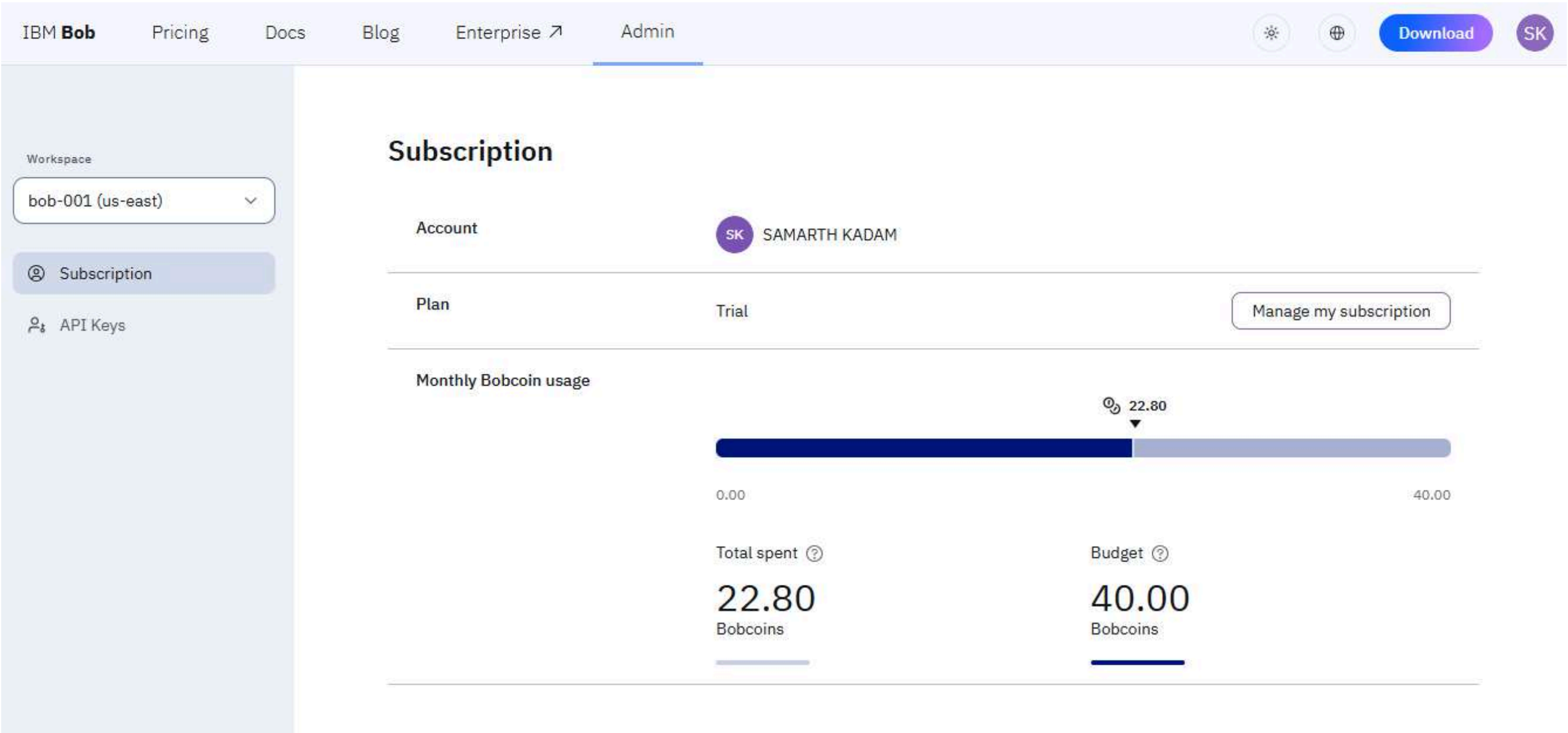
Role of Agentic AI in the solution

- Acts as a Personal Fitness Coach – Provides 24/7 personalized fitness and nutrition guidance.
- Makes Intelligent Decisions – Analyzes user input and recommends suitable workouts, meal plans, or health tips.
- Goal-Oriented Assistance – Tailors advice based on the user's fitness goals, age, BMI, and preferences.
- Context-Aware Interaction – Remembers previous conversations to deliver relevant and consistent responses.
- Ensures Safe Recommendations – Avoids harmful advice and encourages professional consultation when necessary.
- Supports Indian Lifestyle – Suggests culturally relevant Indian diets and fitness practices.

Token Usage



IBM bobToken Usage



Novelty and Uniqueness

- **Indian-First Fitness Coach** – Provides workout and meal recommendations tailored to Indian diets and lifestyles.
- **IBM Granite-Powered AI** – Uses IBM Watsonx.ai for intelligent, safe, and personalized fitness guidance.
- **Customizable AI Agent** – AI behavior (tone, safety rules, and nutrition style) can be easily modified through configuration.
- **All-in-One Platform** – Combines AI chat, BMI calculator, workout planner, meal planner, and habit tracker in a single web application.
- **Context-Aware Agentic AI** – Remembers previous conversations to deliver personalized and goal-oriented responses.
- **Safety-First Design** – Encourages healthy practices and avoids unsafe fitness recommendations.
- **No Installation Required** – Browser-based application with a responsive interface and dark mode support.

Git Hub Link

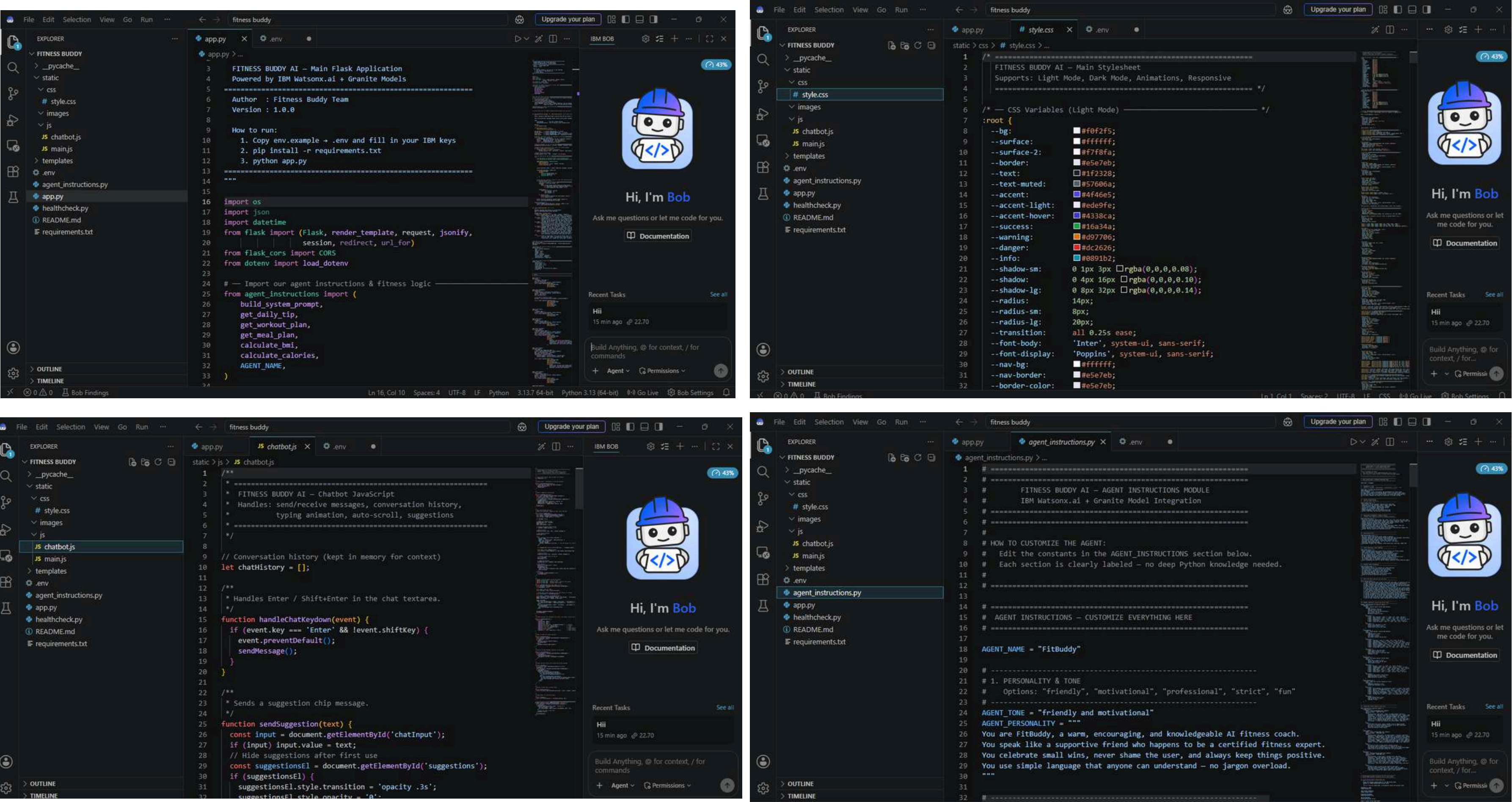
<https://github.com/Samu7709/Fitness-buddy-Project/tree/main>

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Images of project code



Images of project

The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left displays the project structure for 'FITNESS BUDDY', including folders like __pycache__, static, css, images, js, and templates, and files like agent_instructions.py, app.py, healthcheck.py, README.md, and requirements.txt. The main editor area shows the content of agent_instructions.py, which contains comments and instructions for running the application. The terminal panel at the bottom shows the command 'python app.py' being executed, resulting in the application running on http://localhost:5000.

```
1 # =====
2 # =====
3 #     FITNESS BUDDY AI - AGENT INSTRUCTIONS MODULE
4 #     IBM Watsonx.ai + Granite Model Integration
5 # =====
6 # =====
7 #
8 # HOW TO CUSTOMIZE THE AGENT:
9 #   Edit the constants in the AGENT_INSTRUCTIONS section below.
10 #   Each section is clearly labeled - no deep Python knowledge needed.
11 #
```

```
PS C:\Users\lango\OneDrive\Desktop\fitness buddy> python app.py

[FITNESS BUDDY AI] Running on http://localhost:5000
[Agent] FitBuddy (IBM Granite)
[Debug] True

* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.115.102.16:5000
Press CTRL+C to quit
* Restarting with stat

[FITNESS BUDDY AI] Running on http://localhost:5000
[Agent] FitBuddy (IBM Granite)
[Debug] True

* Debugger is active!
```

The screenshot shows the terminal window of the Visual Studio Code editor. It displays the output of the application, including the command 'python app.py' being executed, the application running on http://localhost:5000, and the debugger being active. The terminal also shows a warning message about using a development server in production and a message about a detected change in the app.py file, triggering a reload.

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[FITNESS BUDDY AI] Running on http://localhost:5000
[Agent] FitBuddy (IBM Granite)
[Debug] True

* Debugger is active!
* Debugger PIN: 129-868-790

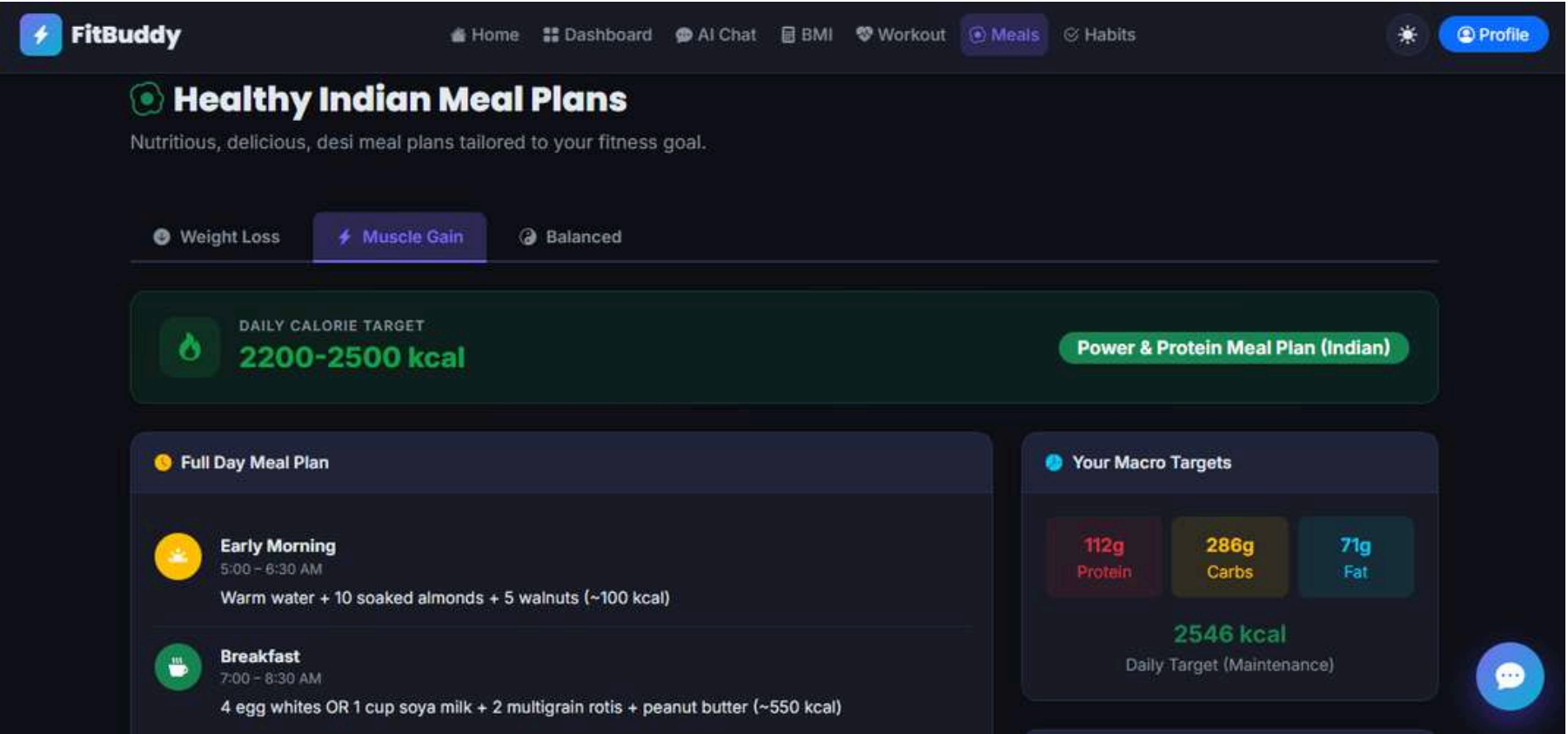
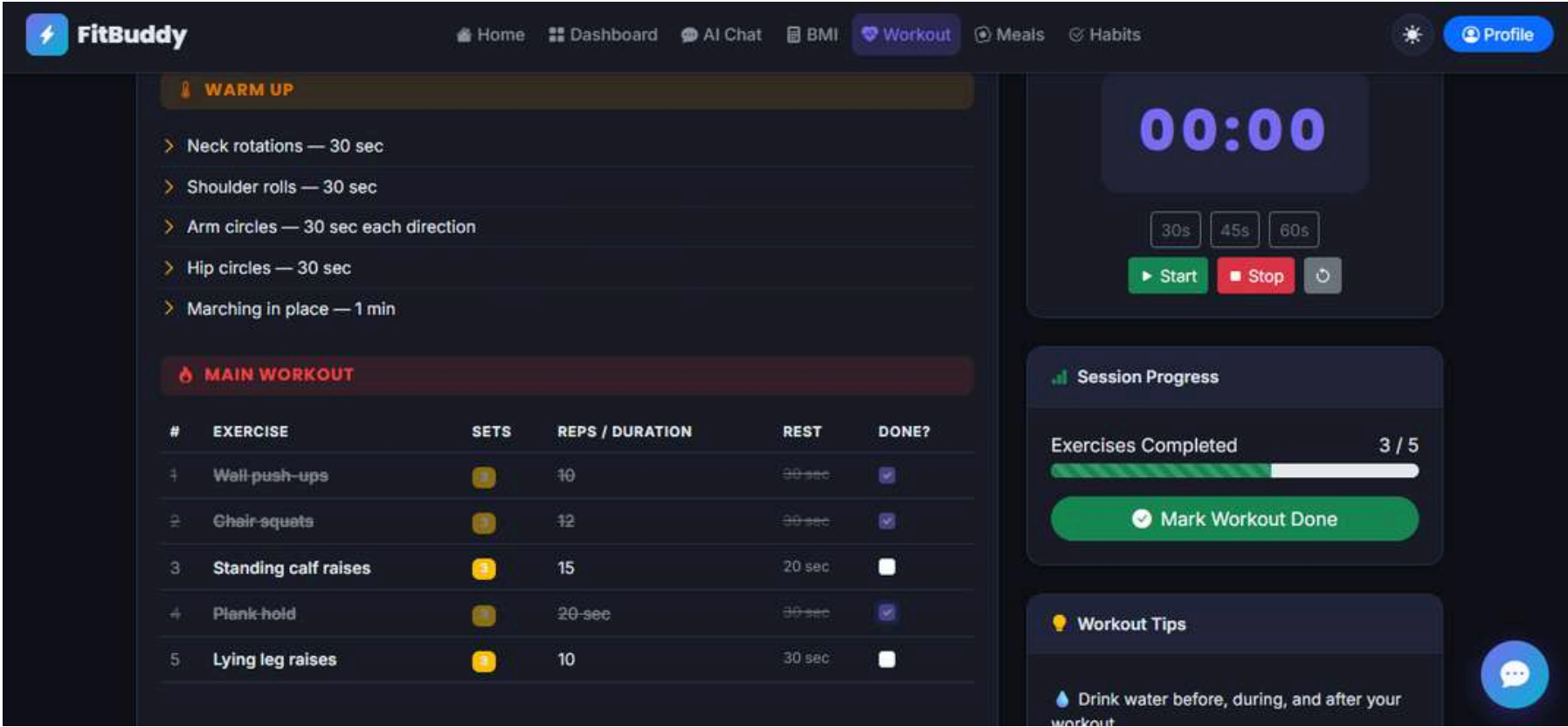
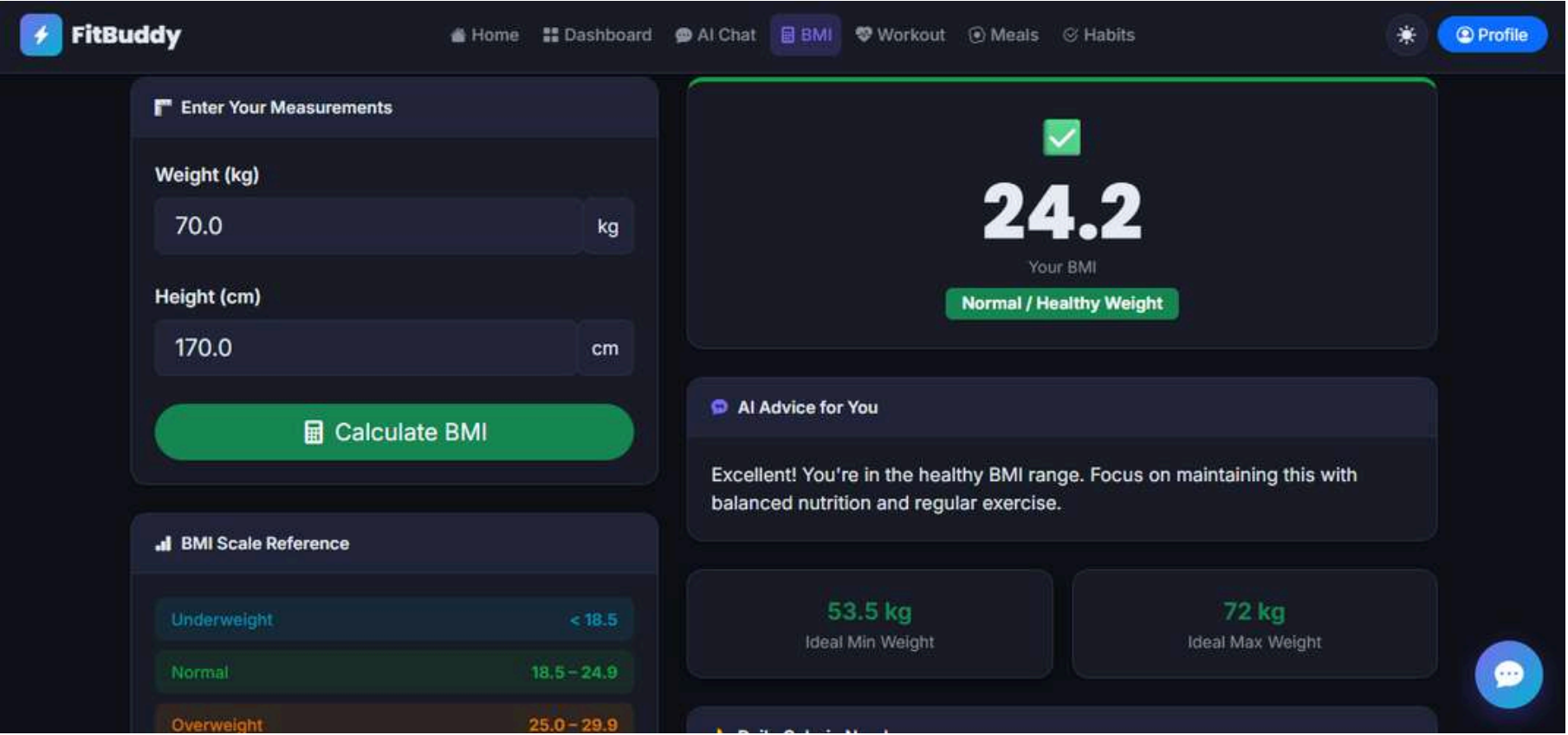
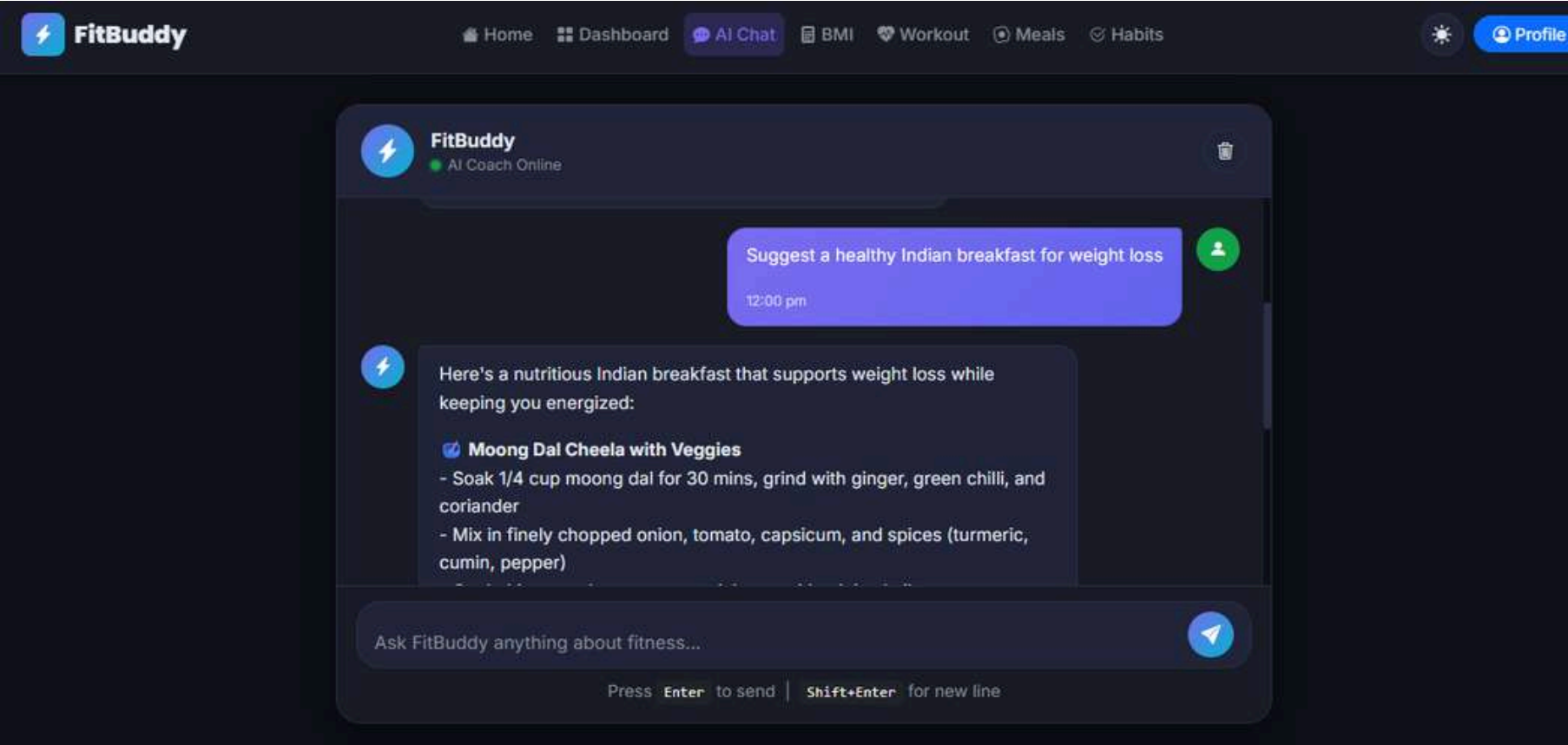
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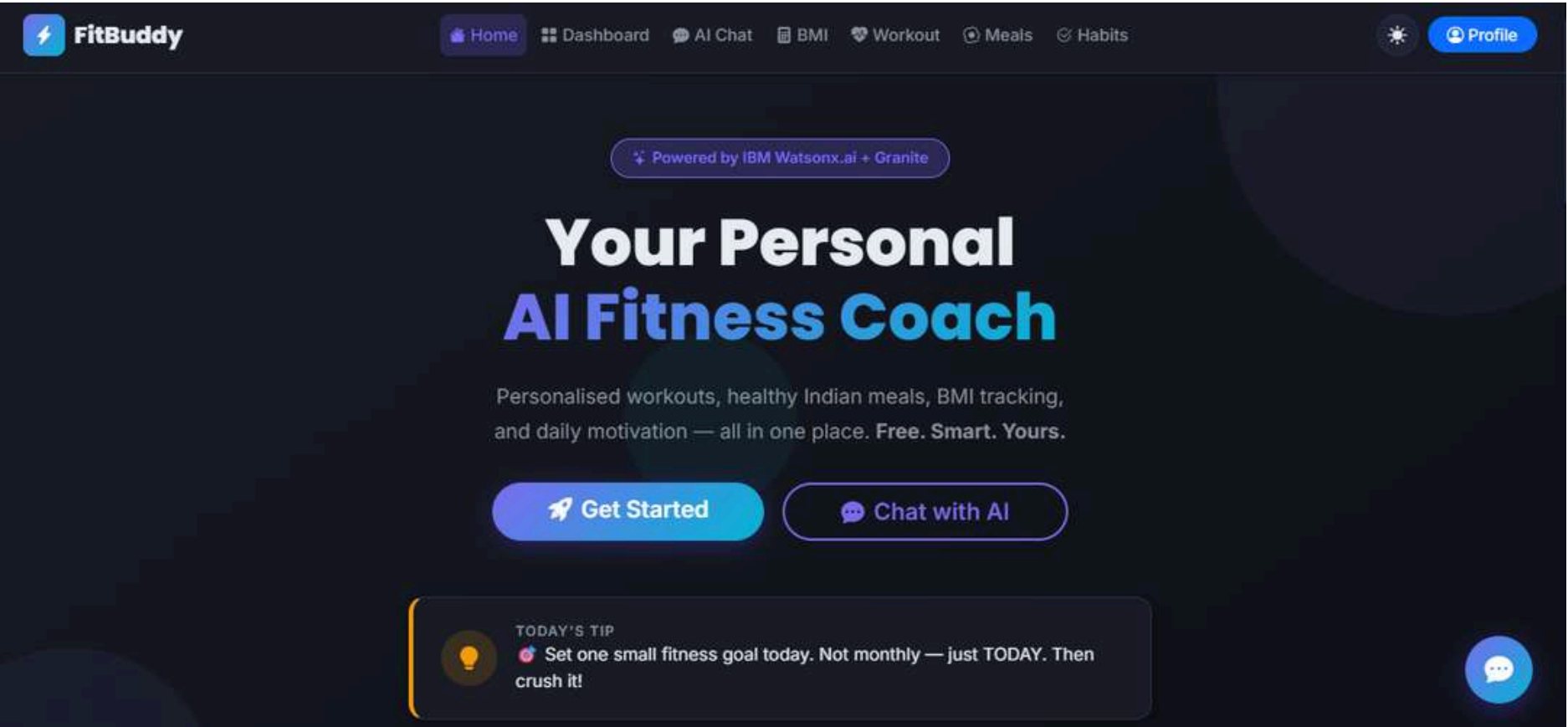
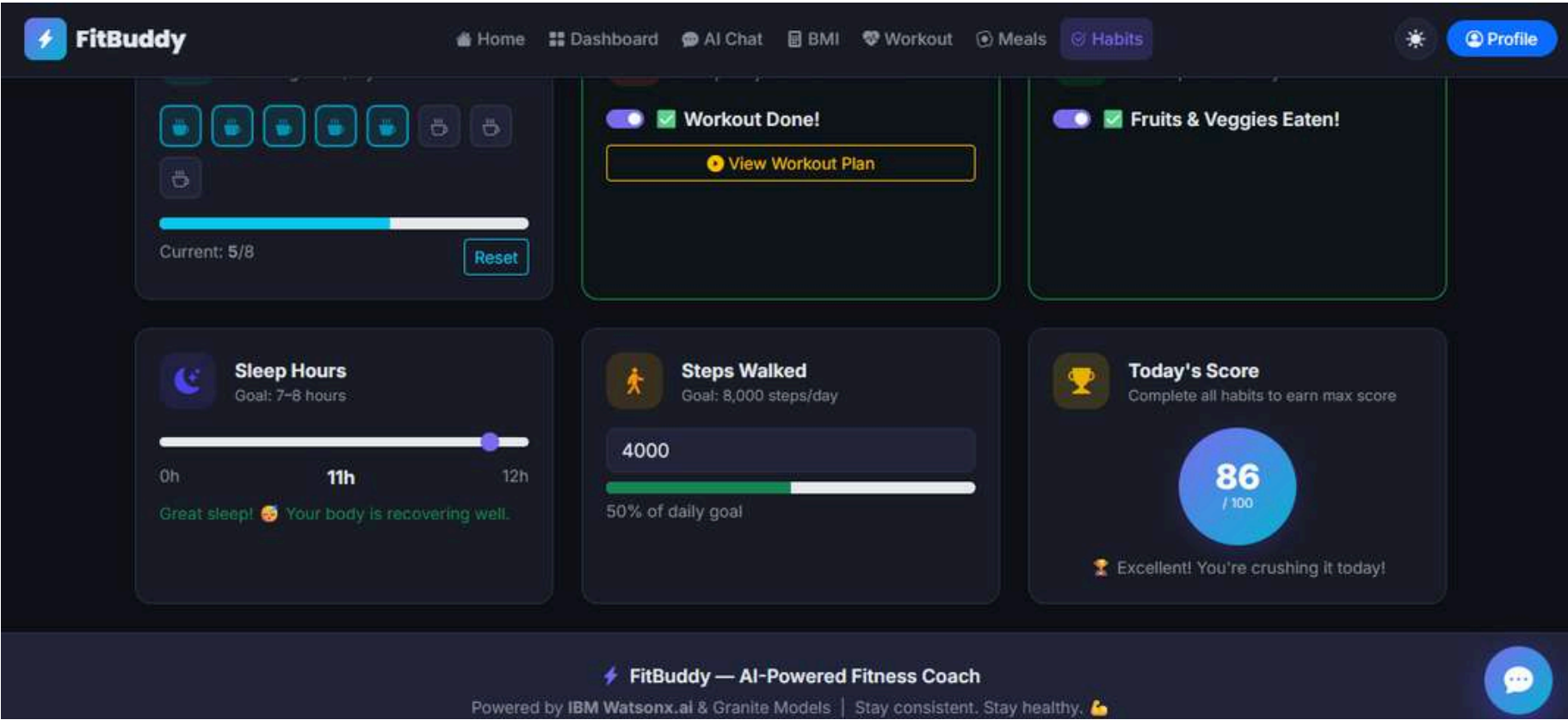
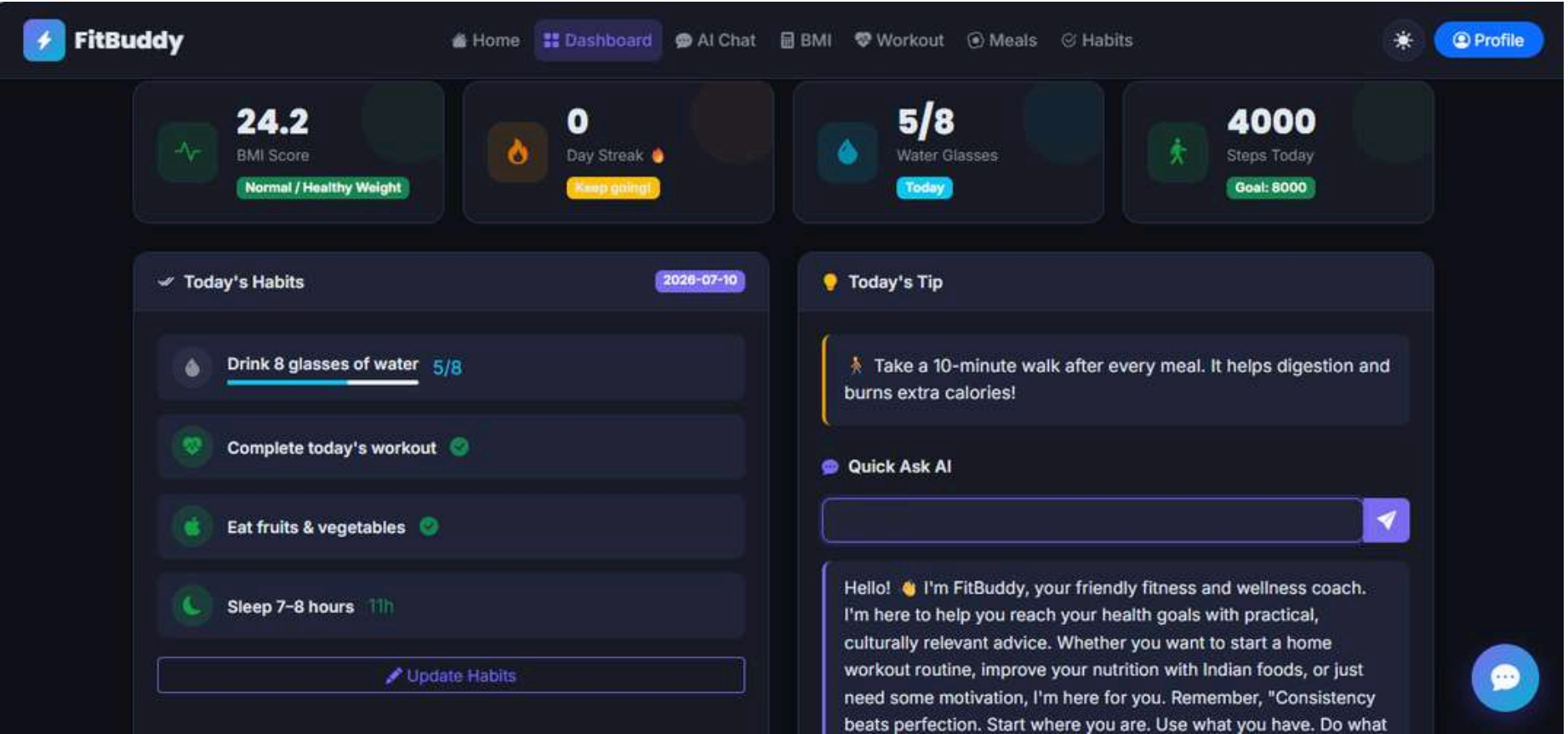
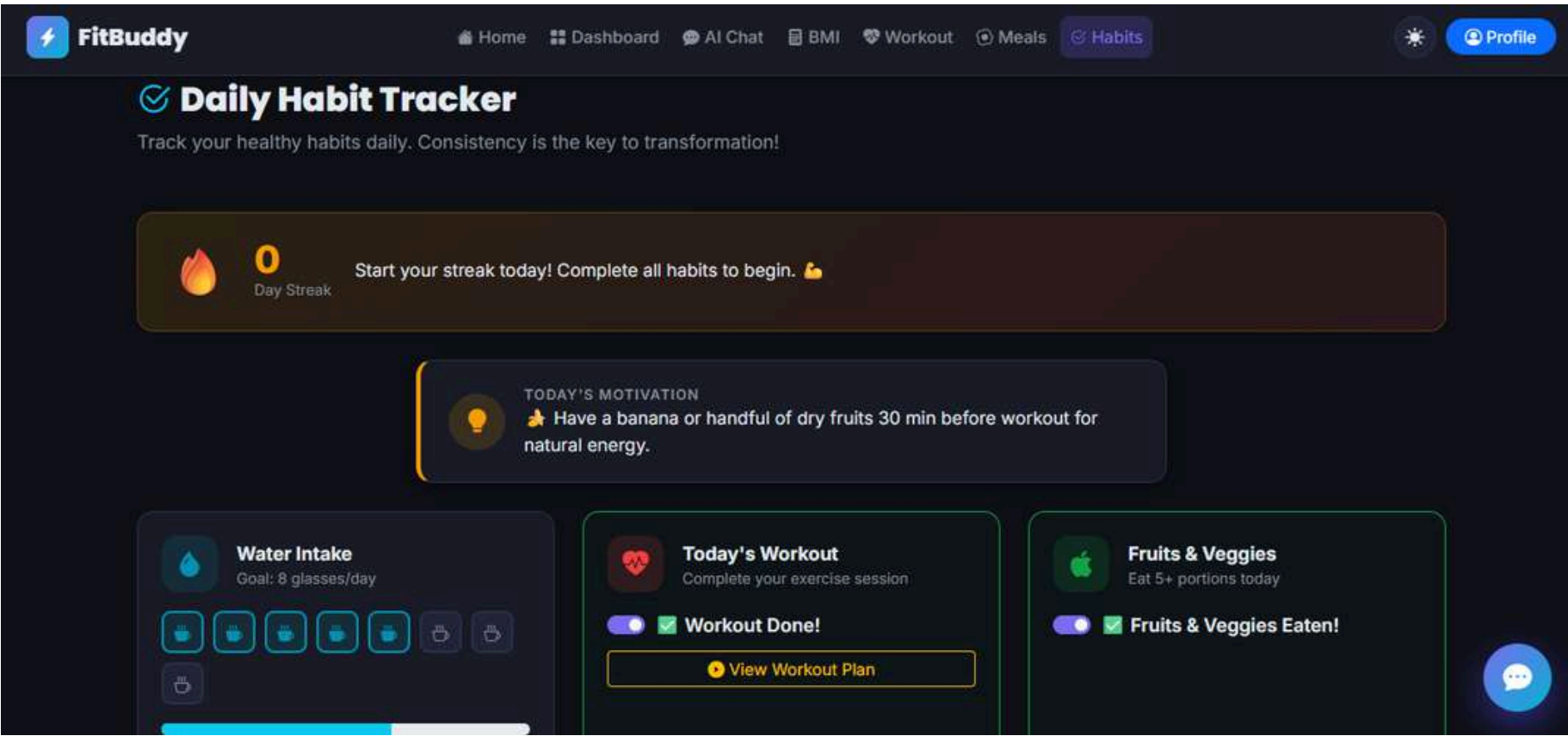
Future Scope

- Mobile Application – Develop Android and iOS versions for easier access.
- Wearable Integration – Connect with smartwatches and fitness bands to track real-time health data.
- Advanced AI Nutrition – Generate personalized meal plans based on medical conditions and dietary preferences.
- Progress Analytics – Provide detailed reports, graphs, and AI-powered fitness insights.
- Exercise Demonstrations – Add video tutorials and AI-based posture correction.
- Multi-language Support – Support regional languages to reach a wider audience.
- Community Features – Introduce fitness challenges, leaderboards, and social sharing for motivation.
- Healthcare Integration – Enable secure sharing of health reports with doctors and fitness professionals.

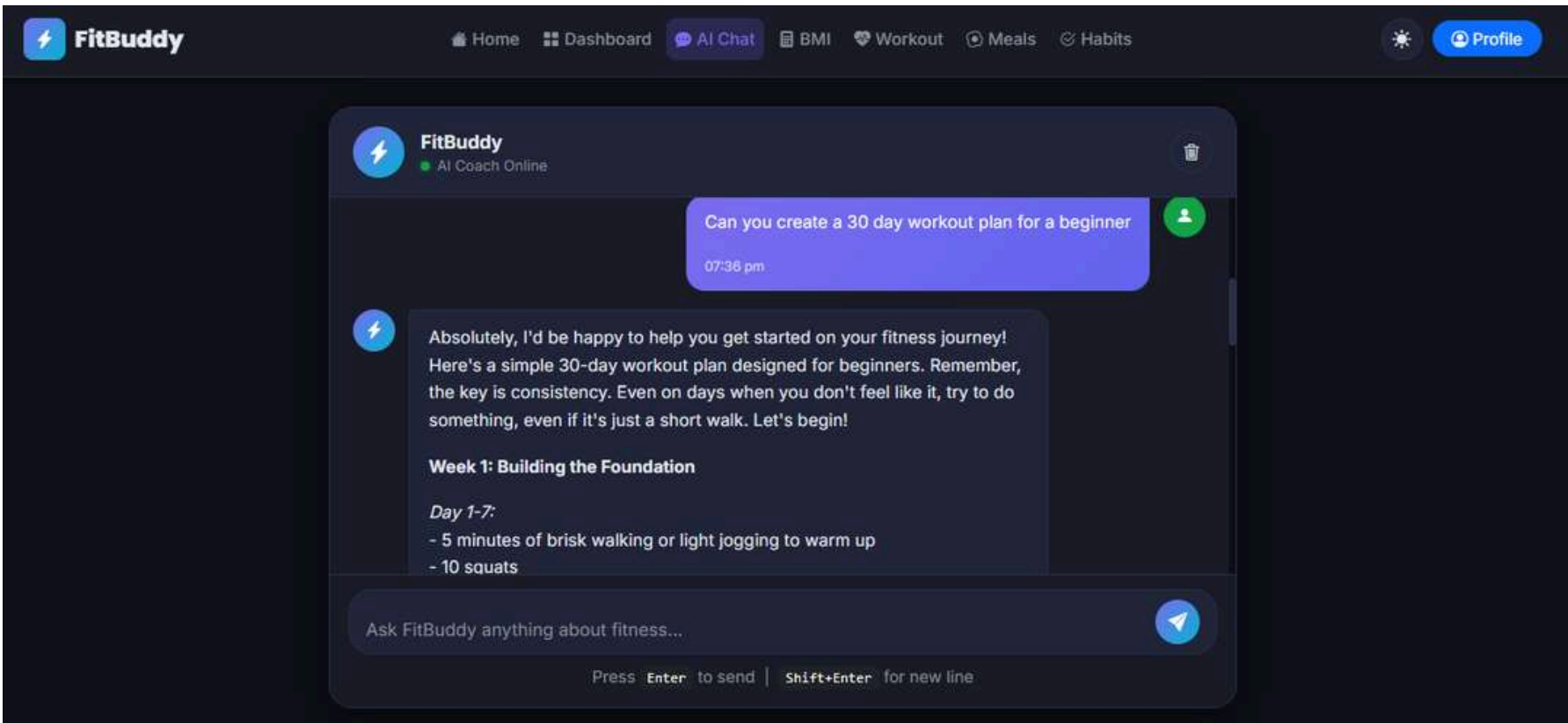
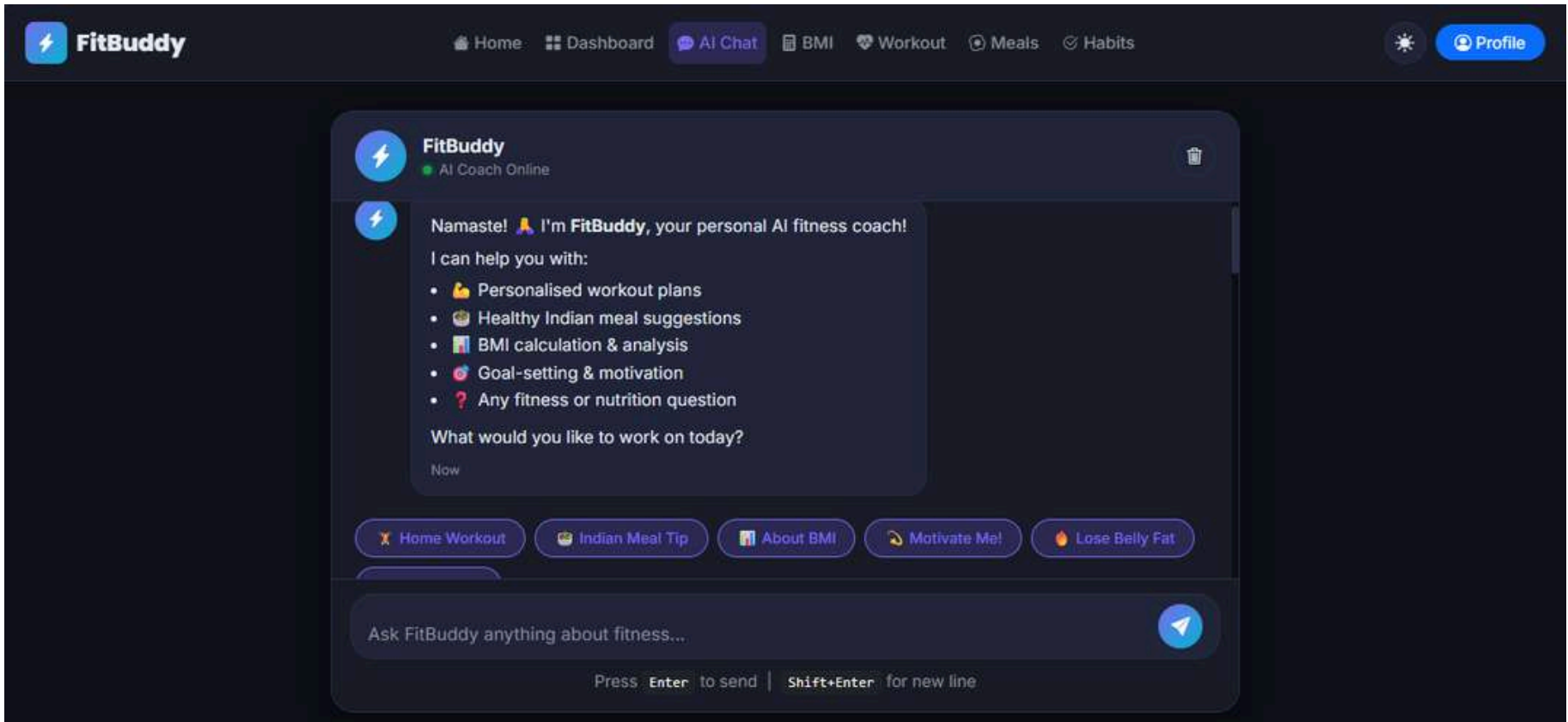
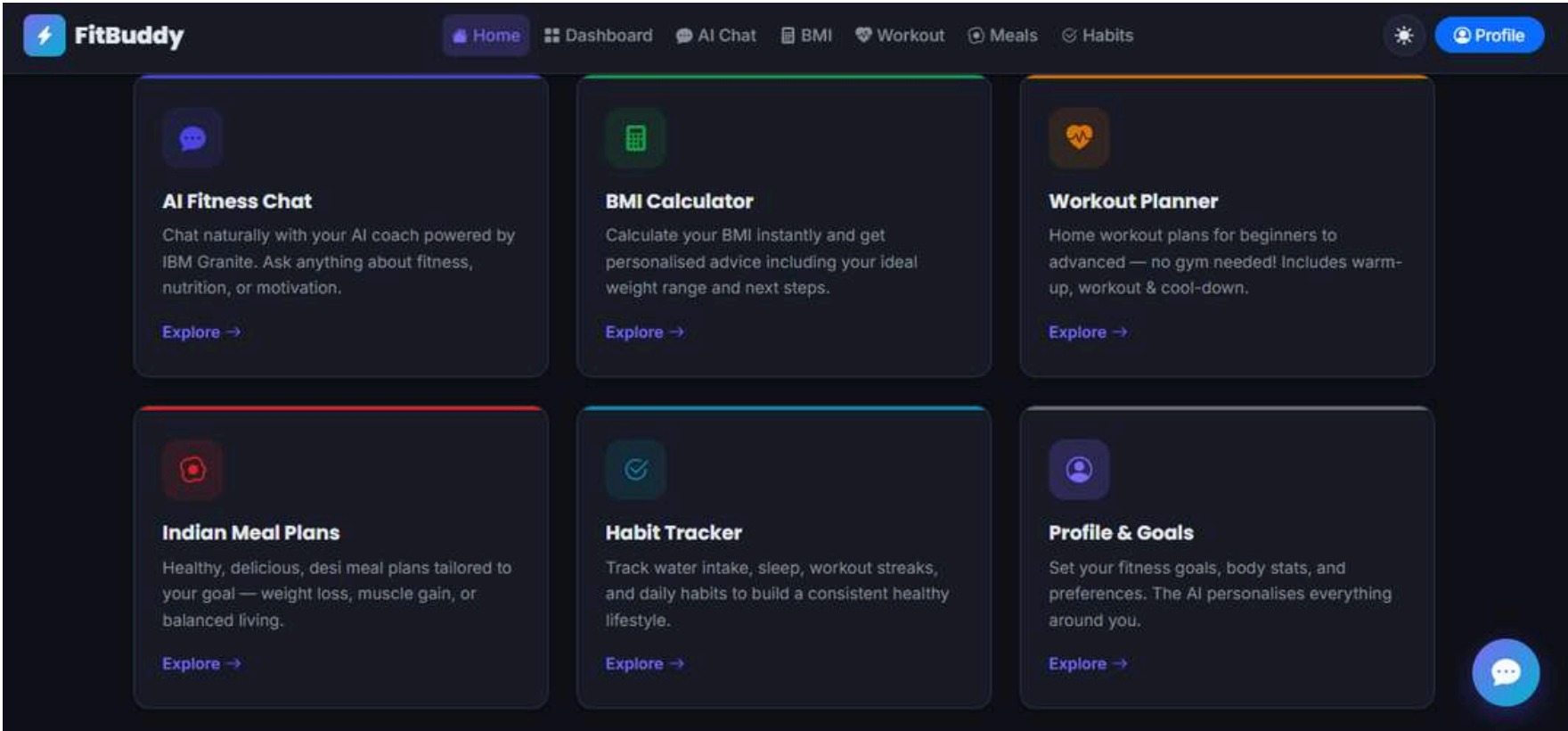
Images of project output



Images of project output



Images of project output



Certificates Earned:



Certificates Earned:



Certificates Earned:



Certificates Earned:

IBM SkillsBuild

Completion Certificate



This certificate is presented to

SAMARTH KADAM

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

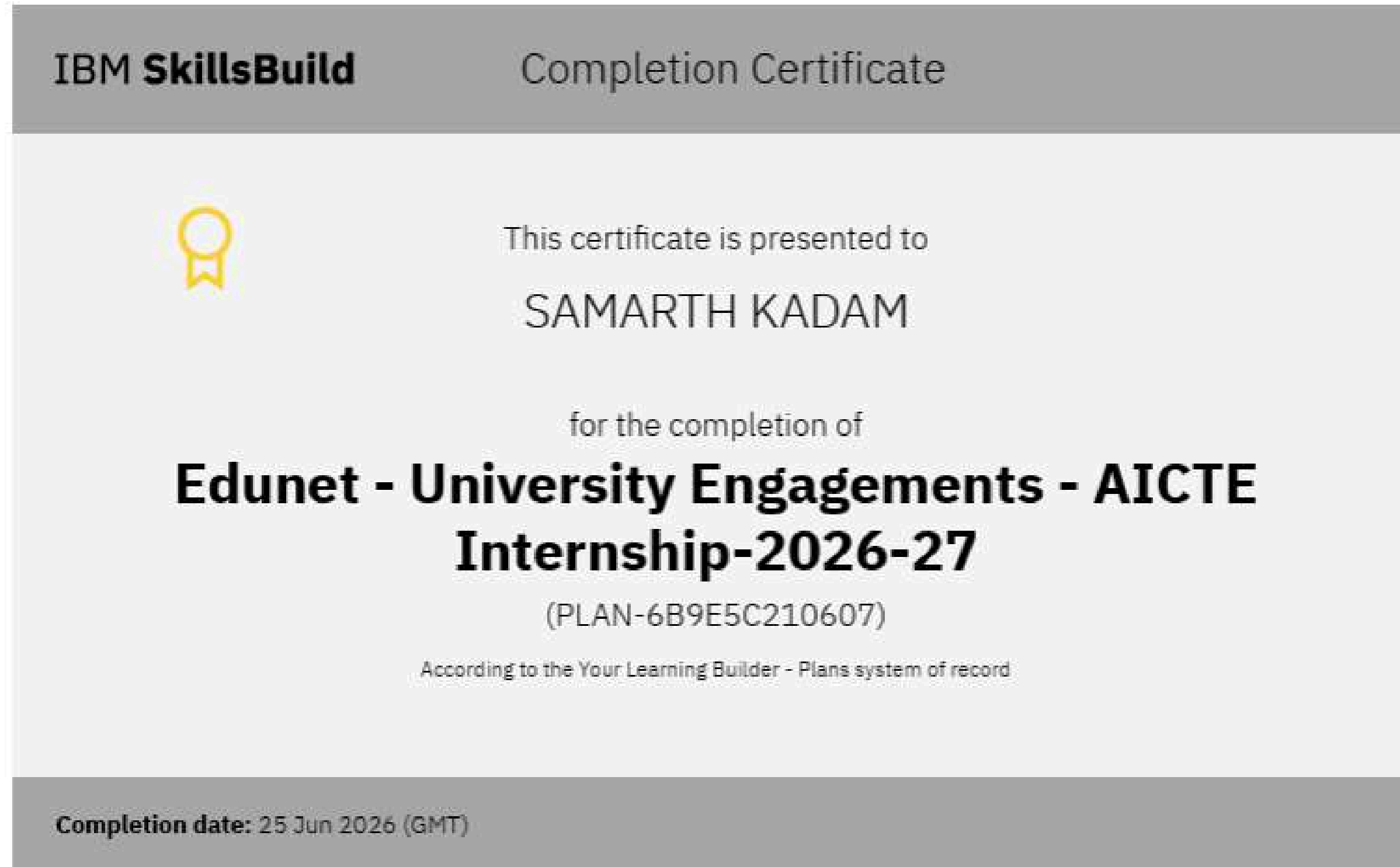
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According to the Adobe Learning Manager system of record

Completion date: 25 Jun 2026 (GMT)

Learning hours: 30 mins

Certificates Earned:



Thank You !